

LAB 06

lab task 1:

```
#include<iostream>
using namespace std;
class Person{
public:
    string name;
    int age;
    void displayInfo(){
        cout<<"Name: "<<name<<"\nAge: "<<age<<endl;
    }
};
class Student:public Person{
public:
    int rollNum;
    void displayStudent(){
        cout<<"Name: "<<name<<"\nAge: "<<age<<endl;
        cout<<"Roll no: "<<rollNum<<endl;
    }
};
int main(){
    Student s1;
    s1.name="Ali";s1.age=19;s1.rollNum=123;
    s1.displayStudent();
}
```

output:

```
cd "/home/rumaisa-abbasi/Desktop/LABS
• rumaisa-abbasi@abdullah-muaz-ThinkPad
/home/rumaisa-abbasi/Desktop/LABS/Lab
Name: Ali
Age: 19
Roll no: 123
• rumaisa-abbasi@abdullah-muaz-ThinkPad
```

Home Task 1:

```
#include<iostream>
using namespace std;
class Vehicle{
public:
    string brand;
    int speed;
    void showVehicle(){
        cout<<"brand: "<<brand<<"\nspeed: "<<speed<<endl;
    }
};
class Car:public Vehicle{
public:
    string fuelType;
    void showCarDetails(){
        cout<<"brand: "<<brand<<"\nspeed: "<<speed<<endl;
        cout<<"fuel type: "<<fuelType<<endl;
    }
};
int main(){
    Car c1;
    c1.brand="Toyota";c1.speed=150;c1.fuelType="petrol";
    c1.showCarDetails();
}
```

Output:

```
cd "/home/rumaisa-abba
• rumaisa-abbasi@abdulla
/home/rumaisa-abbasi/D
brand: Toyota
speed: 150
fuel type: petrol
rumaisa-abbasi@abdulla
```

Lab task 2:

Output:

```
#include<iostream>
using namespace std;
class Employee{
public:
    int ID;
    int salary;
    void showEmployee(){ cout<<"ID: "
    <<ID<<"\nsalary: "<<salary<<endl;}
};
class Manager:public Employee{
public:
    int bonus;
    void showManager(){ cout<<"Salary + Bonus: "
    <<(salary+bonus)<<endl;}
};
int main(){
    Manager m1;
    m1.ID=123;m1.salary=150000;m1.bonus=3000;
    m1.showManager();
}
```

```
cd "/home/rumaisa-abbasi/Desktop/LABS/Lab 06/"
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LABS/Lab 06$ g++ 06.cpp -o 06.exe
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LABS/Lab 06$ ./06.exe
ID: 123
Salary + Bonus: 153000
```

Home task 2:

Output:

```
#include<iostream>
using namespace std;
class Shape{
public:
    int radius;
};
class Circle:public Shape{
public:
    int area(){return (3.14*radius*radius);}
};
int main(){
    Circle c;c.radius=2;
    cout<<"Area of circle: "<<c.area()<<endl;
}
```

```
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LABS/Lab 06$ g++ 06.cpp -o 06.exe
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LABS/Lab 06$ ./06.exe
Area of circle: 12
```

Lab task 3:

Output:

```
#include<iostream>
using namespace std;
class Write{
public:
    void writes(){cout<<"He writes.."<<endl;}
};
class Speak{
public:
    void speaks(){cout<<"He speaks.."<<endl;}
};
class Author:public Write,public Speak{
};
int main(){
    Author a;
    a.writes();a.speaks();
}
```

```
rumaisa-abbasi@abdullah-r
/home/rumaisa-abbasi/Desk
He writes..
He speaks..
rumaisa-abbasi@abdullah-r
```

Home task 3:

Output:

```
#include<iostream>
using namespace std;
class Driver {
public:
    void drive() { cout << "Driving the vehicle..."
    << endl; }
};
class Mechanic {
public:
    void repair(){cout<<"Repairing the vehicle..."
    << endl; }
};
class Technician:public Driver,public Mechanic {
};
int main() {
    Technician t;
    t.drive();
    t.repair();
    return 0;
}
```

```
rumaisa-abbasi@abdullah-mu
/home/rumaisa-abbasi/Deskt
Driving the vehicle...
Repairing the vehicle...
rumaisa-abbasi@abdullah-mu
```

Lab task 4:

Output:

```
#include<iostream>
using namespace std;
class Person {
protected:
    string name;
public:
    void setName(string n) {name = n;}
};
class Employee : public Person {
protected:
    double salary;
public:
    void setSalary(double s) {salary = s;}
};
class Developer : public Employee {
private:
    string language;
public:
    void setLanguage(string l) {language = l;}
    void display() {
        cout << "Name: " << name << endl;
        cout << "Salary: " << salary << endl;
        cout << "Programming Language: " << language << endl;
    }
};
int main() {
    Developer d;
    d.setName("Ali");
    d.setSalary(75000);
    d.setLanguage("C++");
    d.display();
    return 0;
}
```

```
rumaisa-abbasi@abduallah-mu
/home/rumaisa-abbasi/Desktop
Name: Ali
Salary: 75000
Programming Language: C++
rumaisa-abbasi@abduallah-mu
```

Home task 4:

```
#include<iostream>
using namespace std;
class Appliance {
protected: string brand;
public:
    void setBrand(string b) {brand = b;}
    void showBrand() {cout << "Brand: " << brand << endl;}
};
class WashingMachine : public Appliance {
public:
    void wash() {cout << "Washing clothes..." << endl;}
};
class SmartWashingMachine : public WashingMachine {
public:
    void smartControl() {cout << "Controlling washing machine thru mobile app..." << endl;}
};
int main() {
    SmartWashingMachine swm;
    swm.setBrand("Samsung");
    swm.showBrand();
    swm.wash();
    swm.smartControl();
    return 0;
}
```

Output:

```
/home/rumaisa-abbasi/Desktop/LABS/Lab 06/"H4
Brand: Samsung
Washing clothes...
Controlling washing machine thru mobile app..
rumaisa-abbasi@abdullah-muaz-ThinkPad-l390:~/
```


Lab task 5:

```
#include<iostream>
using namespace std;
class Account {
protected:
    int accountNumber;
public:
    void setAccountNumber(int accNo) {accountNumber = accNo;}
    void showAccount() {cout << "Account Number: " << accountNumber << endl;}
};
class SavingsAccount : public Account {
public:
    void calculateInterest(double balance) {
        double interest = balance * 0.05;    // 5% interest
        cout << "Interest: " << interest << endl;
    }
};
class CurrentAccount : public Account {
public:
    void calculateOverdraft(double amount) {cout << "Overdraft Facility Used: " << amount << endl;}
};
int main() {
    SavingsAccount s;
    CurrentAccount c;
    s.setAccountNumber(101);
    c.setAccountNumber(202);
    cout << "Savings Account Details:" << endl;
    s.showAccount();
    s.calculateInterest(10000);
    cout << "\nCurrent Account Details:" << endl;
    c.showAccount();
    c.calculateOverdraft(5000);
    return 0;
}
```

Output:

```
● <esktop/LABS/Lab 06/" && g++ L5.cpp -o L5
Savings Account Details:
Account Number: 101
Interest: 500

Current Account Details:
Account Number: 202
Overdraft Facility Used: 5000
```

Home Task 5:

```
#include<iostream>
using namespace std;
class Media {
protected:
    string title;
public:
    void setTitle(string t) {title = t;}
    void showTitle() {cout << "Title: " << title << endl;}
};
class Book : public Media {
public:
    void showAuthor(string author) { cout << "Author: " << author << endl;}
};
class Video : public Media {
public:
    void showDuration(int minutes) {cout << "Duration: " << minutes << " minutes" << endl;}
};
class Audio : public Media {
public:
    void showSinger(string singer) {cout << "Singer: " << singer << endl;}
};
int main() {
    Book b;
    Video v;
    Audio a;
    b.setTitle("C++ Programming");
    b.showTitle();           // Shared feature
    b.showAuthor("Bjarne");  // Specific feature
    v.setTitle("OOP Tutorial");
    v.showTitle();           // Shared feature
    v.showDuration(45);      // Specific feature
    a.setTitle("Classical Music");
    a.showTitle();           // Shared feature
    a.showSinger("Ali");     // Specific feature
    return 0;
}
```

Output:

```
/home/rumaisa-abbasi/Desktop/LABS/Lab 06/"H5
Title: C++ Programming
Author: Bjarne
Title: OOP Tutorial
Duration: 45 minutes
Title: Classical Music
Singer: Ali
rumaisa-abbasi@abdullah-muaz-ThinkPad-I 390: ~/D
```

Lab Task 6:

```
#include<iostream>
using namespace std;
class University{
public:
    void universityInfo() { cout << "University Function" << endl;}
};
class Faculty:public University{
public:
    void facultyInfo() { cout << "Faculty Function" << endl;}
};
class Administration{
public:
    void adminInfo() {cout << "Administration Function" << endl;}
};
class HOD:public Faculty,public Administration{
public:
    void hodInfo() { cout << "HOD Function" << endl;}
};
int main(){
    HOD h;
    h.universityInfo();
    h.facultyInfo();
    h.adminInfo();
    h.hodInfo();
}
```

Output:

```
cd /home/rumaisa-abbasi/Desktop/LABS
rumaisa-abbasi@abdullah-muaz-ThinkPad:~/Desktop/LABS/Lab
University Function
Faculty Function
Administration Function
HOD Function
rumaisa-abbasi@abdullah-muaz-ThinkPad
```


Home task 6:

```
#include<iostream>
using namespace std;
class Device{
public:
    void powerOn(){cout<<"Device Power ON"<<endl;}
};
class Computer:public Device{
public:
    void processData(){cout<<"Computer Processing Data"<<endl;}
};
class Network{
public:
    void connectNetwork(){cout<<"Connected to Network"<<endl;}
};
class Server:public Computer,public Network{
public:
    void hostWebsite(){cout<<"Server Hosting Website"<<endl;}
};
int main(){
    Server s;
    s.powerOn();
    s.processData();
    s.connectNetwork();
    s.hostWebsite();
    return 0;
}
```

Output:

```
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/
/home/rumaisa-abbasi/Desktop/LABS/Lab 06/"H6
Device Power ON
Computer Processing Data
Connected to Network
Server Hosting Website
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/
```

Lab Task 7:

```
#include<iostream>
using namespace std;
class Base{
public:
    int publicVar=1;
protected:
    int protectedVar=2;
private:
    int privateVar=3;
};
class PublicDerived:public Base{
public:
    void show(){
        cout<<"PublicDerived publicVar: "<<publicVar<<endl;
        cout<<"PublicDerived protectedVar: "<<protectedVar<<endl;
        // cout<<privateVar; // Not accessible
    }
};
class ProtectedDerived:protected Base{
public:
    void show(){
        cout<<"ProtectedDerived publicVar: "<<publicVar<<endl;
        cout<<"ProtectedDerived protectedVar: "<<protectedVar<<endl;
        // cout<<privateVar; // Not accessible
    }
};
class PrivateDerived:private Base{
public:
    void show(){
        cout<<"PrivateDerived publicVar: "<<publicVar<<endl;
        cout<<"PrivateDerived protectedVar: "<<protectedVar<<endl;
        // cout<<privateVar; // Not accessible
    }
};
int main(){
    PublicDerived pd;
    pd.show();
    cout<<"Access publicVar through PublicDerived: "<<pd.publicVar<<endl;
    // cout<<pd.protectedVar; Not accessible
    // cout<<pd.privateVar; Not accessible
    cout<<endl;
    ProtectedDerived protd;
    protd.show();
    // cout<<protd.publicVar; Not accessible (became protected)
    cout<<endl;
    PrivateDerived privd;
    privd.show();
    // cout<<privd.publicVar; Not accessible (became private)
    return 0;
}
```

Output:

```
rumaisa-abbasi@abduallah-muaz-ThinkPad-L390:~/Desktop/LABS/La
&& g++ L7.cpp -o L7 && "/home/rumaisa-abbasi/Desktop/LABS/La
PublicDerived publicVar: 1
PublicDerived protectedVar: 2
Access publicVar through PublicDerived: 1

ProtectedDerived publicVar: 1
ProtectedDerived protectedVar: 2

PrivateDerived publicVar: 1
PrivateDerived protectedVar: 2
rumaisa-abbasi@abduallah-muaz-ThinkPad-L390:~/Desktop/LABS/La
```

Home Task 7:

```
#include<iostream>
using namespace std;
class BankAccount{
public:
    int balance=10000;
    void showBalance(){cout<<"Balance: "<<balance<<endl;}
};
class SavingsAccount:private BankAccount{
public:
    void display(){
        cout<<"Accessing inside derived class:"<<endl;
        cout<<"Balance: "<<balance<<endl;
        showBalance();
    }
};
int main(){
    SavingsAccount s;
    s.display();    // Accessible (through public function)
    // s.balance;    // cant do
    // s.showBalance(); // cant do
    return 0;
}
```

Output:

```
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~$ g++ 07.cpp -std=c++11
/home/rumaisa-abbasi/Desktop/LABS/Lab 06/"H
Accessing inside derived class:
Balance: 10000
Balance: 10000
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~$
```

Lab Task 8:

```
#include<iostream>
using namespace std;
class Person{
public:
    void show(){cout<<"Person Function"<<endl;}
};
class Student : public Person {
};
class Teacher : public Person {
};
class TA:public Student,public Teacher {
};
int main(){
    TA t;
    t.Student::show(); // Resolves ambiguity temporarily
    t.Teacher::show();
    return 0;
}
```

Output:

```
/home/rumaisa-abbasi/
Person Function
Person Function
rumaisa-abbasi@abdulla
```

Home Task 8:

Output:

```
#include<iostream>
using namespace std;
class Machine {
public:
    void show() { cout << "Machine" << endl; }
};
class Printer : public Machine {
};
class Scanner : public Machine {
};
class AllInOne : public Printer, public Scanner {
};
int main() {
    AllInOne obj;
    // obj.show(); // Ambiguous: compiler doesn't know which
    // | Machine::show() to call
    obj.Printer::show(); // Works: call Printer's path
    obj.Scanner::show(); // Works: call Scanner's path
    return 0;
}
```

```
rumaisa-abbasi@abdu...
/home/umaisa-abbasi/D
Machine
Machine
```

Lab Task 9:

```
#include<iostream>
using namespace std;
class Machine {
public:
    void show() { cout << "Machine" << endl; }
};
class Printer : virtual public Machine {
public:
    Machine* getBasePtr() { return this; }
};
class Scanner : virtual public Machine {
public:
    Machine* getBasePtr() { return this; }
};
class AllInOne : public Printer, public Scanner {
public:
    Machine* getBasePtr() { return this; }
};
int main() {
    AllInOne obj;
    obj.show(); // No ambiguity
    cout << "Printer->Machine: " << obj.Printer::getBasePtr() << endl;
    cout << "Scanner->Machine: " << obj.Scanner::getBasePtr() << endl;
    cout << "AllInOne->Machine: " << obj.getBasePtr() << endl;
    return 0;
}
```

Output:

```
ca /home/umaisa-abbasi/Desktop/L
• rumaisa-abbasi@abdu...
/home/umaisa-abbasi/Desktop/LABS/
Machine
Printer->Machine: 0x7ffdec002270
Scanner->Machine: 0x7ffdec002270
AllInOne->Machine: 0x7ffdec002270
rumaisa-abbasi@abdu... Think
```


Home task 9:

```
#include<iostream>
using namespace std;
class Device {
public:
    void showDevice() { cout << "Device" << endl; }
};
class Camera : virtual public Device {
public:
    void takePhoto() { cout << "Taking Photo" << endl; }
};
class Phone : virtual public Device {
public:
    void makeCall() { cout << "Making Call" << endl; }
};
class SmartPhone : public Camera, public Phone {
public:
    void browseInternet() { cout << "Browsing Internet" << endl; }
};
int main() {
    SmartPhone sp;
    sp.showDevice();      // from Device
    sp.takePhoto();       // from Camera
    sp.makeCall();        // from Phone
    sp.browseInternet();  // own function
    return 0;
}
```

Output:

```
rumaisa-abbasi@abduallah-m
/home/rumaisa-abbasi/Desk
Device
Taking Photo
Making Call
Browsing Internet
rumaisa-abbasi@abduallah-m
```


Final:

Output:

```
#include<iostream>
#include<string>
using namespace std;
class Person {
public:
    string name;
protected:
    int age;
private:
    int privateID;
public:
    Person(string n="", int a=0, int id=0) : name(n), age(a), privateID(id) {}
    void showPublic() { cout << "Name: " << name << endl; }
    void showProtected() { cout << "Age: " << age << endl; }
    void showPrivate() { cout << "ID: " << privateID << endl; }
};
class Teacher : virtual public Person {
public:
    Teacher() {}
    void teach() { cout << name << " is teaching." << endl; }
};
class Student : virtual public Person {
public:
    Student() {}
    void study() { cout << name << " is studying." << endl; }
};
class TeachingAssistant : public Teacher, public Student {
public:
    TeachingAssistant() {}
    void assist() { cout << name << " is assisting in class." << endl; }
};
class ExamAssistant : public TeachingAssistant {
public:
    ExamAssistant(string n, int a, int id): Person(n, a, id) {}
    void conductExam() { cout << name << " is conducting an exam." << endl; }
};
int main() {
    ExamAssistant ta("Ali", 22, 1001);
    cout << "Access Base Class Members" << endl;
    ta.showPublic();
    ta.showProtected();
    ta.showPrivate();
    cout << "\nAccess Derived Class Functions" << endl;
    ta.teach();
    ta.study();
    ta.assist();
    ta.conductExam();
}
```

Access Base Class Members

Name: Ali

Age: 22

ID: 1001

Access Derived Class Functions

Ali is teaching.

Ali is studying.

Ali is assisting in class.

Ali is conducting an exam.

o rumaisa-abbasi@abdullah-muaz-Think